Huawei Access Node KR-ACC-41 Routing Protocol Flap SOP

Node: KR-ACC-41 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4470

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on Huawei Access Node KR-ACC-41 can be attributed to various underlying factors such as misconfigured routing protocols, hardware malfunction, or software issues. In a multi-access, multi-protocol environment like KR Puram Access, the interaction between BGP and OSPF/ISIS can lead to unforeseen consequences like routing oscillations, which may cause the node to flap or oscillate between different routing states. Furthermore, hardware-related issues like faulty transceivers, incorrect cabling, or overheating can also precipitate this failure domain. As the Huawei Access Node KR-ACC-41 relies on its robust routing and switching capabilities to ensure efficient data forwarding, any disruption in these services can lead to a service-impacting event. Therefore, timely detection and resolution of the root cause are crucial to prevent further outages and ensure network stability.

Pre-requisite Checks

- Ensure remote hands are dispatched to KR Puram Access for physical access and assistance.

- Verify that redundant links are up and stable, ensuring minimal impact on network services.

- Check the node's environmental conditions, including temperature, humidity, and power supply, to rule out any hardware-related issues.

- Review the node's routing configurations to ensure consistency and accuracy.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing protocol	High
Faulty transceiver or incorrect cabling	Medium
Software issue or bug	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Routing table information, including route counts and AS path lengths, Description: This command isolates the BGP routing table and provides information about route counts, AS path lengths, and other relevant details. It helps identify any anomalies or discrepancies in the routing information.

Command: display ospf neighbor, Expected Output: List of OSPF neighboring nodes and their statuses, Description: This command
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```
displays the list of OSPF neighboring nodes and their statuses, helping to identify any adjacency loss or issues.

Command: display isis database, Expected Output: ISIS LSP database information, including LSP counts and sequence numbers, Description: This command isolates the ISIS LSP database and provides information about LSP counts, sequence numbers, and other relevant details. It helps identify any anomalies or discrepancies in the LSP database.
```

Resolution Steps

Step 1: Step: Verify and correct BGP routing configuration, Command: system-view bgp 64497 undo network 10.0.0.0/8, Description: This step involves verifying and correcting the BGP routing configuration to ensure consistency and accuracy. The command `undo network 10.0.0.0/8` is used to remove any incorrect or unnecessary network statements.

Step 2: Step: Replace faulty transceiver or correct cabling, Command: replace transceiver module, Description: This step involves replacing the faulty transceiver or correcting the cabling to ensure proper communication between nodes.

Step 3: Step: Update software or patch software issue, Command: system-view software install, Description: This step involves updating or patching the software to resolve any bugs or issues. The command `software install` is used to initiate the software installation process.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Routing table information, including route counts and AS path lengths, Description: This command is used to verify that the BGP routing table is stable and accurate.

Command: display ospf neighbor, Expected Output: List of OSPF neighboring nodes and their statuses, Description: This command is used to verify that the OSPF neighboring nodes are stable and reachable.

Command: display isis database, Expected Output: ISIS LSP database information, including LSP counts and sequence numbers, Description: This command is used to verify that the ISIS LSP database is stable and accurate.
```

Roll-back Procedures

- Step: Restore original BGP routing configuration, Command: system-view bgp 64497 network 10.0.0.0/8, Description: This step involves restoring the original BGP routing configuration to its previous state.

- Step: Revert transceiver module replacement, Command: replace transceiver module revert, Description: This step involves reverting the transceiver module replacement to its previous state.

- Step: Revert software installation, Command: system-view software rollback, Description: This step involves reverting the software installation to its previous state.